

# Requirements

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## Functional Requirements

- The system shall allow gym members to register and log in to their accounts.
- The system shall allow the admin to add, update, and delete members.
- The system shall manage gym subscriptions including start and end dates.
- The system shall allow members to book training sessions (personal or group).
- The system shall allow members to make payments for subscriptions or sessions.
- The system shall allow trainers to view their assigned training sessions.

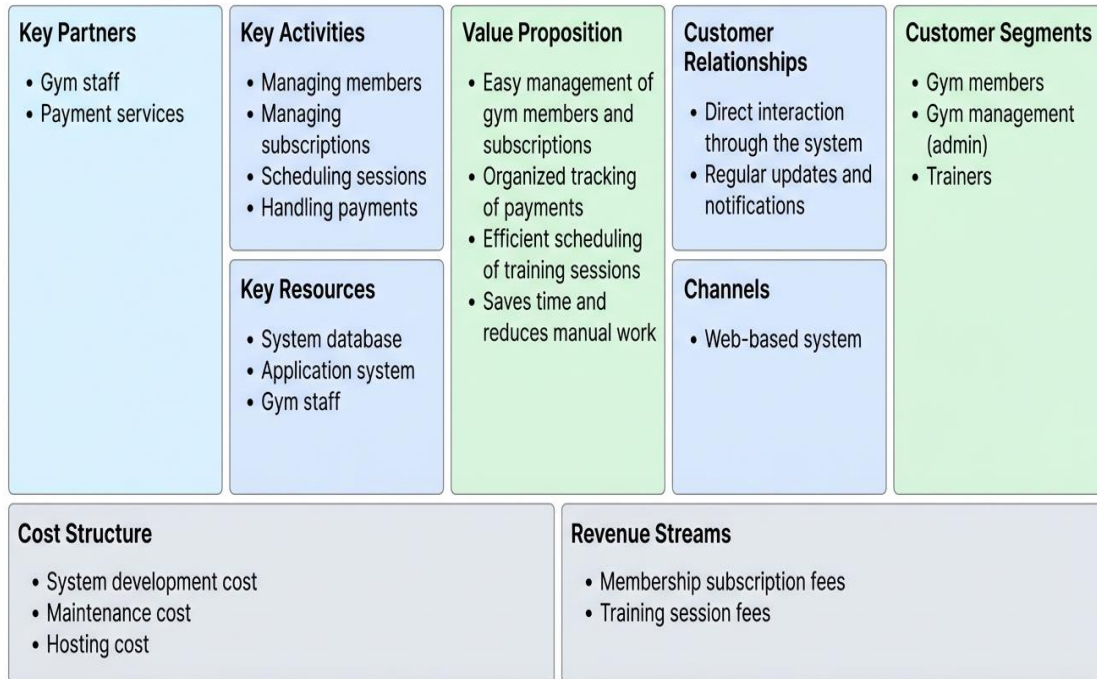
## Non-Functional Requirements

- The system should be easy for gym members and staff to use.
- The system should respond quickly when booking sessions or making payments.
- The system should protect member data and payment information.
- The system should be available during working hours of the gym.
- The system should handle multiple users (members, trainers, admin) at the same time.

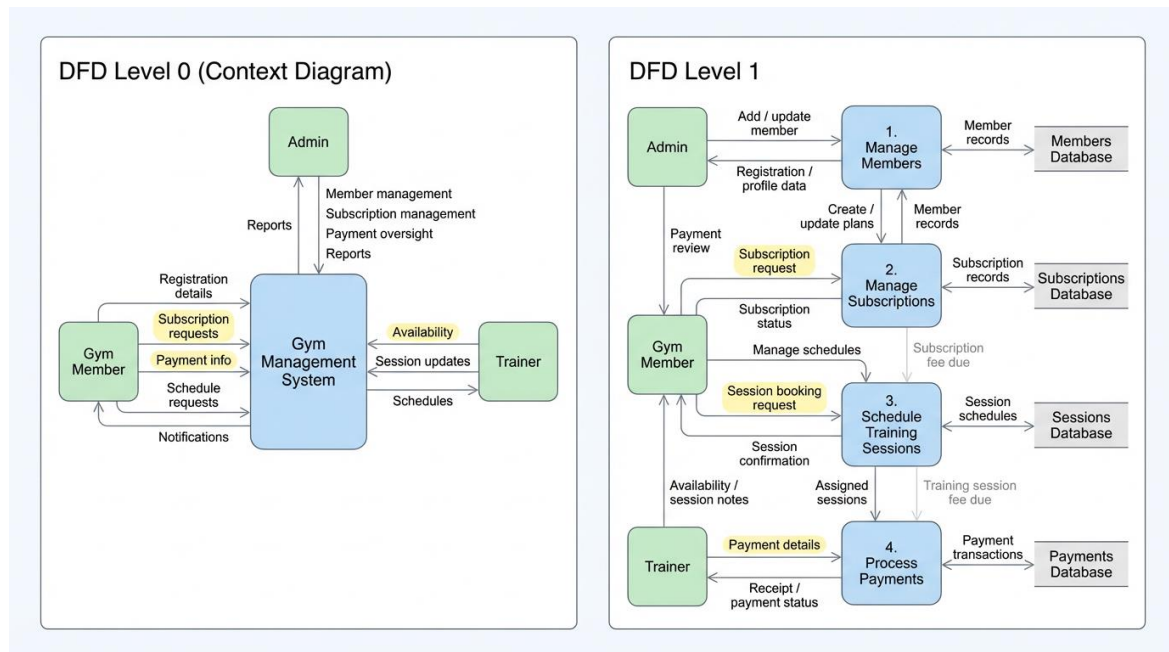
# Business Model Canvas

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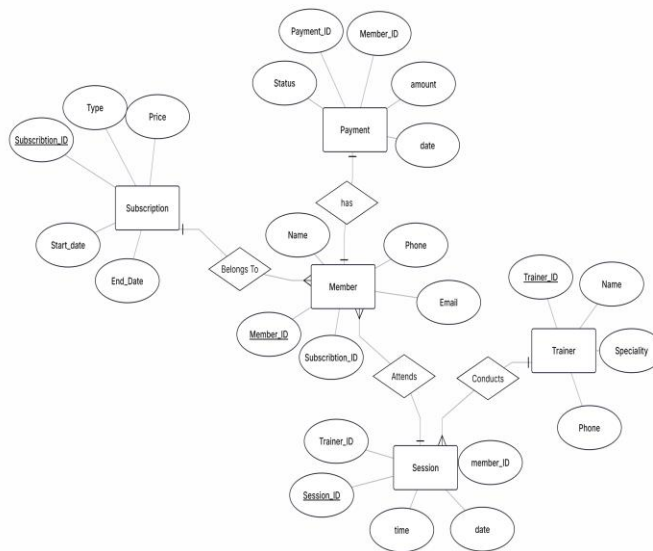
## Business Model Canvas - Gym Management System



# Data Flow Diagram (DFD)

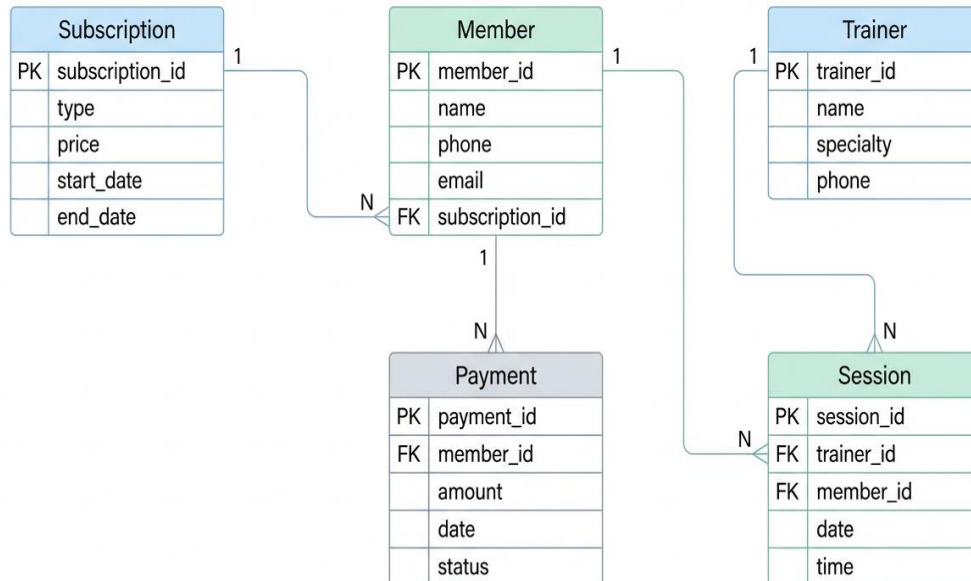


# Entity Relationship Diagram (ERD)

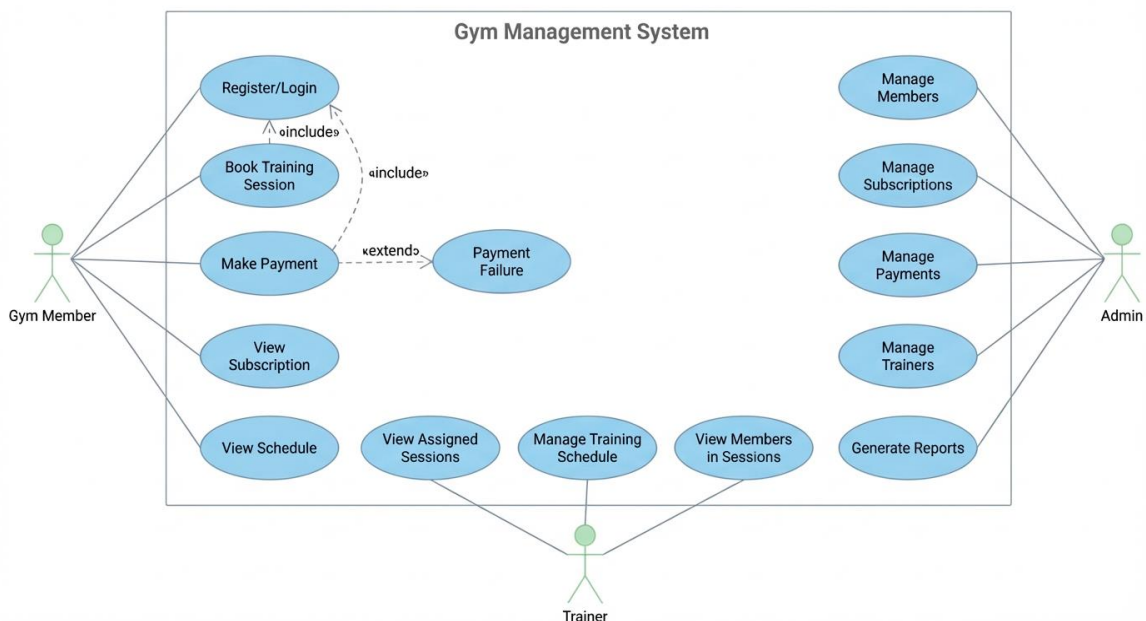


# Data Schema

Database Schema Diagram - Gym Management System



# Use Case diagram



# Use Case Scenarios

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## Use Case 1: Booking a Training Session

Actor: Gym Member

Description:

This scenario explains how a gym member books a training session using the system. The member can choose between different types of sessions such as personal training or group sessions.

Preconditions:

- The member must have an account
- The member must have an active subscription

Main Flow:

1. The member logs into the system
2. The system displays available training sessions (e.g., cardio session, personal training)
3. The member selects a session
4. The system checks if the session is available
5. If available, the system confirms the booking
6. The session is added to the member's schedule

Alternative Flow:

- If the selected session is full, the system shows a message that the session is not available

Postconditions:

- The booking is saved and the member can view it in their schedule

## Use Case 2: Making a Payment

Actor: Gym Member

Description:

This scenario explains how a gym member pays for a subscription or a training session using the system.

Preconditions:

- The member must be logged into the system
- The member must have a valid subscription or a pending payment

Main Flow:

1. The member logs into the system

2. The member goes to the payment section
3. The system shows the amount that needs to be paid
4. The member confirms the payment
5. The system processes the payment
6. The system updates the payment status
7. The member receives a confirmation

Alternative Flow:

- If the payment fails, the system shows an error message and asks the member to try again

Postconditions:

- The payment is recorded in the system
- The member's account is updated successfully